Functional and differential equations – 2

Functional-differential equations

Homework

13 сентября 2026 года

Theoretical minimum. These equations connect f and f' at different points, and the first move is always to gain regularity: an integral over a moving window gives one derivative, and iterating gives all of them. A mean value equation is solved by fixing one argument and differentiating in the other, which converts it into an ordinary differential equation; the answer depends on the mean, and for the arithmetic mean it is a quadratic polynomial. A delay equation is solved by $e^{\lambda x}$, but the characteristic equation is transcendental and has infinitely many roots, so "all solutions" is a statement about a series, not a finite basis. Equations with a composition are rigid: one compares monotonicity and growth of the two sides and usually gets a contradiction. Finally, a differentiable solution of a functional equation may be found by differentiating in each variable separately and eliminating.

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1. (RMO) Find all differentiable functions $f\colon [0,\infty) \to [0,\infty)$ with $f(0) = 0$ and $f'(x^2) = f(x)$ for all $x \geqslant 0$.

2. (RMO 2015) Find all differentiable $f\colon \mathbb{R} \to \mathbb{R}$ such that $f'(a) = 0$ for every integer a and $f'(x) = 0$ implies $f(x) = 0$.

3. Prove that there is no differentiable $f\colon \mathbb{R} \to \mathbb{R}$ with $f'(x) = f(f(x))$ and $f(0) > 0$, and compare with the IMC problem on $f'(t) > f(f(t))$ in the homework of Differentiable functions – 1.

4. Find all differentiable f with $f'(x) = f(x+1)$ for all x; describe the solutions of exponential type and show that the characteristic equation $\lambda = e^{\lambda}$ has no real roots.

5. Find all continuous $f\colon \mathbb{R} \to \mathbb{R}$ satisfying $f(x) = \frac{1}{2h}\int_{x-h}^{x+h} f$ for a fixed $h > 0$ and all x, and compare the answer with the mean value property of harmonic functions (Complex analysis – 8).

6. Find all differentiable $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x+y) = f(x)+f(y)+f'(x)f'(y)$ for all x, y.

7. Let f be differentiable with $f(a) - f(b) = (a-b)f'\left(\frac{2ab}{a+b}\right)$ for all positive a, b (the harmonic mean). Find all such f, and explain the general principle: for which means μ does the equation have non-polynomial solutions?

8. Find all C^1 functions with $f'(x) = f(1-x)$ for all $x \in \mathbb{R}$, and all C^2 functions with $f''(x) = f(-x)$.

9. Let f be continuous on $\mathbb{R}$ and satisfy $f(x) = f(2x)$ for all x. Prove that f is constant; then find all differentiable f with $f'(x) = f(2x)$ and $f(0) = 1$, as a power series, and determine its radius of convergence.

10. Prove that a solution of $f'(x) = f(x-1)$ which is bounded on $\mathbb{R}$ and tends to zero at $-\infty$ is identically zero, and explain which property of the characteristic roots is responsible.